

ML-KEM-1024 on CV32E40X / CW305: detailed cycle and TVLA

measurements

companion to oscar_17may26_v1.pdf

(measurement record)

2026-05-17

This document is a measurement journal for `oscar_17may26_v1.tex`. Every cell here is reproducible from the artefact tree containing this document (`cw305_artifacts_mlkem_sw_v1/`) plus its sibling `cw305_artifacts_2026-05-15/` (the ISE configurations referenced for comparison). All paths in this document are relative to the artefact root that contains this file; sibling artefact paths begin with `../`.

1 Platform and silicon target

FPGA	Xilinx XC7A35T-2FTG256 on ChipWhisperer CW305
Core	OpenHW Group CV32E40X, RV32IMC, no ISE-extended ABI
Clock	target 16 MHz (P&R <code>cv32e40x_core.clk</code> reports 20.46 MHz Fmax, PASS at 20 MHz)
SRAM	64 KB at <code>0x10000000..0x1000_FFFF</code> (boot at <code>0x10000080</code> , stack at <code>0x1000FFF0</code>)
RTL provenance	<code>../cw305_artifacts_2026-05-15/rtl/</code> (git <code>d3d21d9</code>) – snapshot of the source artefact’s RTL, used at build time
SoC top crt0	<code>integration/sca_pqc_cw305_top.sv</code> of the parent OSPQC-Pro working tree <code>firmware/src/crt0_mlkem_sw.S</code> – 8 phase markers + <code>csrwi 0x320,0</code> (clears <code>mcountinhibit</code> so <code>mcycle</code> runs)
mlkem-native	<code>pq-code-package/mlkem-native</code> portable-C SCU, no native arith backend
Toolchain	<code>yosys</code> (<code>oss-cad-suite</code>), <code>nextpnr-xilinx</code> , <code>prjxray-db</code> <code>artix7</code> , <code>riscv64-unknown-elf-gcc</code>

2 Measured cycle counts (Section V, Table I in the paper)

2.1 SW-only (this work) – live `mcycle` delta per phase

Phase	Cycles (dec)	Cycles (hex)	k cycles	Time @ 16 MHz	MBOX slot
KeyGen	5 892 477	0x0059E97D	5 892	368.3 ms	MBOX[0]
Encap	6 026 116	0x005BF384	6 026	376.6 ms	MBOX[1]
Decap	7 401 699	0x0070F0E3	7 402	462.6 ms	MBOX[2]
Sentinel	–	0xCAFE0001	–	–	MBOX[3] = PASS (all 5 byte-checks pk/sk)

Bitstream: `bitstreams/sca_pqc_cw305_mlkem_sw_kat_artifactrtl_cyc.bit`
ELF: `bitstreams/firmware_mlkem_sw_kat_artifactrtl_cyc.elf`

2.2 ISE and ISE + DOM (carried over from paper)

The ISE rows in Table I cite Decap = 180 k cyc (naive ISE) and 401 k cyc (ISE + DOM), which were measured by the original authors on the same CV-X-IF ISE platform. KeyGen and Encap for those rows have not been independently re-measured for this measurement journal and are therefore

omitted; if those numbers are needed, they should be read live from the same MBOX [0..2] protocol when the ISE bitstreams are run on the same CW305 board.

2.3 Decap speed-up (vs measured SW-only baseline)

Configuration	Decap (k cyc)	Decap (cyc)	vs. SW
SW-only (this work)	7 402	7 401 699	1.0×
ISE	180	180 000	41.1× (7 401 699 / 180 000)
ISE + DOM	401	401 000	18.5× (7 401 699 / 401 000)

The paper previously cited $\sim 13.9\times$ and $6.2\times$ against an *estimated* ~ 2500 k SW Decap. The measured 7 402 k pushes the speed-up numbers to $\sim 41\times$ (naive ISE) and $\sim 18\times$ (ISE + DOM). The ratio of cost between ISE+DOM and naive ISE is unchanged at $2.23\times$ – that’s a firmware-only swap on byte-identical RTL.

3 TVLA – share-recombine window, Saarinen-PQC pool (Table II in the paper)

3.1 Protocol

Standard	ISO/IEC 17825:2024 (Welch- t , $\geq 4.5\sigma$, $N \geq 10$ k) under Saarinen’s PQC adaptation [1]
Target op	ML-KEM-1024 Decapsulation
Target window	Fujisaki–Okamoto share-recombine: the implicit-rejection comparison $c \stackrel{?}{=} c'$ where the masked plaintext m' is unmasked and the plaintext-check oracle bit transiently exists
Trigger source	MBOX[1] pulse mapped to CW305 TIO4, bracketed around <code>mlk_poly_tomsg</code> in <code>indcpa.c</code>
Pool A	1 fixed valid ciphertext <code>SAARINEN_CT_A</code> (correct under sk_{fixed} , HW = 6294 bits)
Pool B	10-element pool of invalid ciphertexts under sk_{fixed} , each with Hamming weight in $\{6286, \dots, 6304\}$ – within ± 10 bits of HW(A) so byte-load HW cancels in the t -statistic
A/B coin	32-bit Galois LFSR seeded from <code>mcycle</code> , pool label published in MBOX[3] bit 16
Scope	ChipWhisperer-Lite, ADC = $4 \times$ target = 64 MHz, presamples = 250, samples = 6000, gain 25 dB
Post-trace fix	per-trace DC subtract (removes slow PSU / thermal drift)

3.2 Headline numbers

Configuration	N/pool	1st-ord. $ t $	2nd-ord. $ t $	Var ratio $ 1 - r $	Verdict (vs 4.5σ)
SW (no ISE issued)	1 k	4.66σ	3.84σ	0.2696	marginal (preliminary, $N < 10$ k)
SW (no ISE issued)	10 k	13.96σ	3.17σ	0.0701	LEAK (1st-order, broadband)
ISE naive (pos. ctrl)	5 k	71.55σ	53.82σ	0.85	LEAK (positive control)
ISE + DOM (masked)	10 k	3.10σ	4.01σ	0.0784	PASS

3.3 SW-only, N=1k – preliminary sample-level statistics

N_A, N_B	1000, 1000
Sample count T	6000 (presamples 250 + post-trigger 5750)
Drops (firmware $\rightarrow$ host)	3
1st-order $ t $ pre-trigger max	2.96σ (samples $[0, 250)$) – below threshold, sanity OK
1st-order $ t $ post-trigger max	4.66σ at sample 1116 – 0.16σ over threshold
2nd-order $ t $ pre-trigger max	1.82σ
2nd-order $ t $ post-trigger max	3.84σ at sample 1381
Var(A)/Var(B) post-trigger range	0.7790 .. 1.2696 ($ 1 - r _{\max} = 0.2696$)

This $N=1k$ run is preliminary and below the ISO 17825 floor. Its only use in this document is as the N -scaling reference for the ISO-compliant $N=10k$ result that follows.

3.4 SW-only, N=10k – ISO 17825-compliant sample-level statistics

N_A, N_B	
Sample count T	
Drops (firmware $\rightarrow$ host)	
Capture rate	
On-disk size	
1st-order $ t $ pre-trigger max	
1st-order $ t $ post-trigger max	
Samples post-trigger above 4.5σ	
Longest run of consecutive samples above 4.5σ	
2nd-order $ t $ pre-trigger max	
2nd-order $ t $ post-trigger max	3.17σ at sample 2884 – below threshold
Var(A)/Var(B) post-trigger range	0.9299 .. 1.0626 ($ 1 - r _{\max} = 0.0701$) – pools have essentially identical noise

Reading.

- $\sqrt{N}$ scaling holds. The $N=1k$ preliminary 4.66σ projects to $4.66 \cdot \sqrt{10} \approx 14.7 \sigma$ at $N=10k$. Measured: 13.96σ . Within 5 % of projection – so the small- N signal was a real leak, not a statistical sampling artefact.
- Magnitude is $\sim 5\times$ smaller than the ISE-naive positive control projected to the same N ($71.55 \sigma \cdot \sqrt{2} \approx 101 \sigma$ at $N=10k$ from the $N=5k$ reading). The SW Decap takes 7 402 k cycles versus 180 k for the ISE datapath, so the same share-recombine event is diluted across $\sim 41\times$ more cycles on the CV32E40X core. The per-cycle power density on a general-purpose RISC-V core is also lower than on a dedicated PQC datapath. Together these effects predict an order-of-magnitude smaller signal, observed.
- The leak is broadband. 149 isolated samples cross threshold but the ISO 17825 default “run of 5 consecutive samples” flag does not trigger. This is the signature of a secret-dependent operation that runs across many cycles in software rather than firing in a narrow window. It is a genuine first-order leak but a wide-band attacker model (e.g., mean-based DPA over the whole trace) would exploit it, whereas a narrow-band single-sample template attack would gain less.
- Variance ratio is tighter than the masked baseline. SW $|1 - r|_{\max} = 0.0701$ vs. masked ISE 0.0784. The two unmasked pools have nearly identical noise distributions; all the leakage is in the per-sample mean. This is the textbook unmasked first-order DPA target.

3.5 Trace and report file paths

All paths in this table are relative to the artefact root `cw305_artifacts_mlkem_sw_v1/` (so each row is reachable by `cd cw305_artifacts_mlkem_sw_v1 && cat <row>`). The sibling ISE artefact `cw305_artifacts_2026_05_15/` is reached via `../`.

SW N=1k setA, setB	<code>traces/mlkem_sw_widetrig_n1k/cw305_mlkem_sw_widetrig_artifactrtl_set{A,B</code>
SW N=1k 1o, 2o, vr	<code>reports/mlkem_sw_widetrig_n1k/{tvla...1o_t,tvla...2o_t,varratio...}.npz</code>
SW N=10k setA	<code>traces/mlkem_sw_widetrig_n10k/cw305_mlkem_sw_widetrig_artifactrtl_n10k.se</code> (120 MB)
SW N=10k setB	<code>traces/mlkem_sw_widetrig_n10k/cw305_mlkem_sw_widetrig_artifactrtl_n10k.se</code> (120 MB)
SW N=10k 1o, 2o, vr	<code>reports/mlkem_sw_widetrig_n10k/{tvla...n10k_1o_t,</code> <code>tvla...n10k_2o_t, varratio...n10k}.npz</code>
ISE naive 5k 1o	<code>../cw305_artifacts_2026_05_15/reports/share_recombine_n5k/tvla_share_rec</code>
ISE masked 10k 1o	<code>../cw305_artifacts_2026_05_15/reports/share_recombine_n10k/tvla_share_rec</code>

4 Re-derive any number in this document

```
# SW N=10k 1st-order (ISO 17825-compliant). Run from inside this artefact root.
python3 -c "
import numpy as np
d = np.load('reports/mlkem_sw_widetrig_n10k/'
            'tvla_mlkem_sw_widetrig_n10k_1o_t.npz')
t = d['t']
print(f'post-trigger max|t| = {abs(t[250:]).max():.2f}')
print(f'N_fixed = {int(d["n_fixed"])}', N_random = {int(d["n_random"])}')
"
```

The same recipe with `tvla...2o_t.npz` returns the 2nd-order $|t|$, and `varratio...npz["ratio"]` returns the variance-ratio curve from which $\max|1-r|$ is computed.

5 Pending

- ISE naive and ISE+DOM KeyGen / Encap live cycle counts – not re-measured for this journal; the paper’s Decap-only row remains the authoritative speed-up reference. Adding these is one probe run away on the existing `cw305_artifacts_2026_05_15/bitstreams/sca_pqc_cw305_saarinen_naive` and the `*masked_v2*.bit` when the per-call MBOX protocol there is rerun.
- 2nd-order positive control at $N \geq 10^5$ per pool (Saarinen-PQC recommended for the masked side) – future work.
- Pre-silicon (RTeAAL) cross-layer analysis on the SW row – the RTeAAL L1/L3 maps in the main paper are for the ISE configurations only. Re-running them with the SW firmware would let us check whether the broadband nature of the SW first-order leak shows up as a wide tile spread at L3, as the diluted-power-density hypothesis predicts.

References

- [1] M.-J. O. Saarinen, “Side-channel test methodology for ML-KEM/ML-DSA,” HOST WiP, 2025.